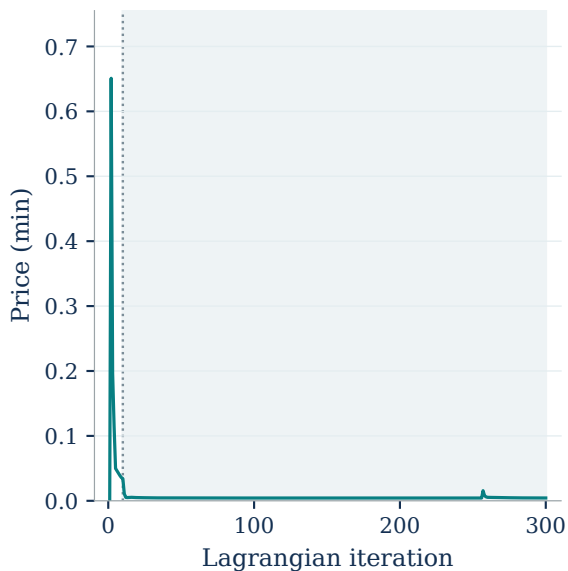
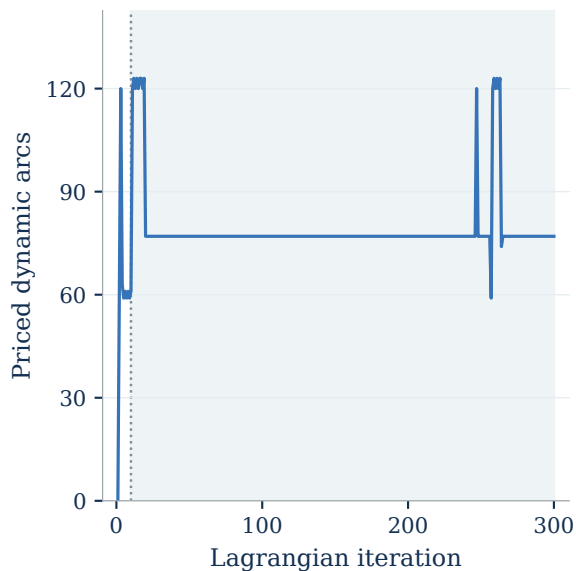


Lagrangian prices and relaxed capacity violations

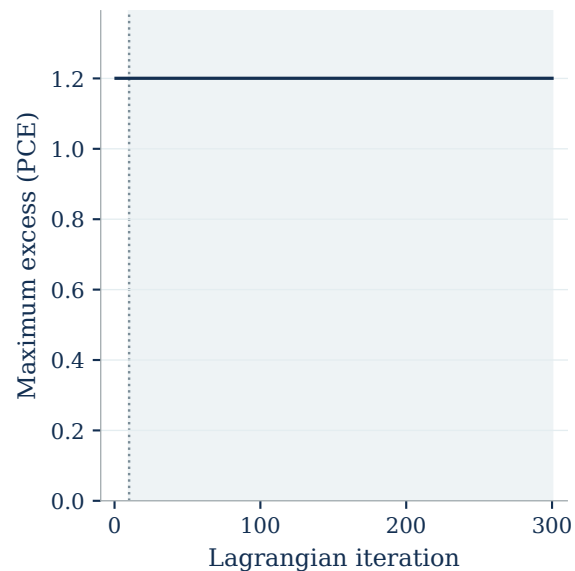
a Maximum capacity multiplier



b Positive-price dynamic arcs



c Relaxed capacity excess



Dotted line: first formal certificate at iteration 10. Shading: subsequent diagnostic rounds.